The Fine Structure Constant as a Spectral Coincidence on the Three-Sphere

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We record a numerical observation. The number $K = \pi(42 + \pi^2/6 - 1/40) = 137.036064$ lies within 4.8×10^{-7} (relative) of the inverse fine structure constant α^{-1}, and the smallest positive root of the cubic $\alpha^3 - K\alpha + 1 = 0$ lies within 8.8×10^{-8} of α (CODATA 2018). The three ingredients are spectral data of the unit three-sphere S^3: $B = 42$ is the degeneracy-weighted trace of the $SO(3)$ Casimir $l(l+1)$ over the scalar S^3 harmonics of the first three shells, the cutoff being the only one at which $B/N = \dim S^3 = 3$; $F = \zeta(2) = \pi^2/6$ is the Dirichlet series of the shell degeneracy n^2 at $s = 4$; and $\Delta = 1/40$ is the reciprocal of a Dirac degeneracy on S^3. The combination rule $K = \pi(B+F-\Delta)$ is an Observation, not a derivation: its weights, signs, prefactor and the cubic are chosen, and we list which steps are computed and which are chosen. In a search over $N = 1.92 \times 10^9$ formulas built from ingredients of the same kind, ten match α more closely ($p = 5.2 \times 10^{-9}$ per formula), so the expected chance count is $Np \approx 10$: within that search the formula is not unusual. The search was run after the formula was found and does not correct for that. Twelve mechanisms proposed to generate all three ingredients from one spectral construction were tested and eliminated; we tabulate them.

INTRODUCTION

The fine structure constant $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137.036$ is an unexplained input to the Standard Model [1]. Numerical formulas for it have a long and unsuccessful history [2, 3]; Wyler's symmetric-space expression [8] reached comparable precision and was criticized for lacking any mechanism connecting the mathematics to the coupling [9].

This paper records one such formula and states what is and is not known about it. Its ingredients come from the unit three-sphere S^3, onto which Fock's 1935 stereographic projection maps the hydrogen bound-state problem [4, 6]. The GeoVac label lattice has as nodes the labels (n, l, m) of the S^3 harmonics, on which the Laplace–Beltrami operator acts as the diagonal $SO(4)$ Casimir $n^2 - 1$ (the lattice's edge Laplacian does not converge to that operator, Paper 7). Nothing in the formula uses the edge structure; it uses only the labels, their degeneracies, and two spectral series.

We state the status at the outset. The combination rule $K = \pi(B + F - \Delta)$ is a numerical Observation. Each ingredient has a closed form, but the way they are combined is chosen, and a look-elsewhere count shows that a search of this size is expected to produce about ten matches this close by chance. An earlier version of this work [7] used a "symplectic impedance" ratio with one fitted parameter (6.2×10^{-5} relative error); it is superseded.

The sections below define the three ingredients and the cutoff, state the formula and the cubic, give the look-elsewhere count, separate computed from chosen steps, record where each ingredient lives and the twelve eliminated mechanisms, and state the scope of any Hopf-fibration reading of the formula.

THE THREE INGREDIENTS

B: a Casimir trace over the scalar S^3 harmonics

Each shell n of the scalar harmonics on S^3 carries the (j, j) representation of $SO(4) \cong (SU(2) \times SU(2))/\mathbb{Z}_2$, $j = (n-1)/2$, which branches under the rotation subgroup $SO(3)$ into $l = 0, \ldots, n-1$. Define the degeneracy-weighted trace of the $SO(3)$ Casimir over the first $n_{\max}$ shells,

$$B(n_{\max}) = \sum_{n=1}^{n_{\max}} \sum_{l=0}^{n-1} (2l+1)\, l(l+1). \tag{1}$$

At $n_{\max} = 3$,

$$\begin{aligned} B(3) &= \underbrace{0}_{(1,0)} + \underbrace{0}_{(2,0)} + \underbrace{6}_{(2,1)} + \underbrace{0}_{(3,0)} + \underbrace{6}_{(3,1)} + \underbrace{30}_{(3,2)} \\ &= 42, \end{aligned} \tag{2}$$

with subscripts the (n, l) pairs.

B is an S^3 quantity, not an S^2 one. The Laplacian on S^2 has eigenvalues $l(l+1)$ with multiplicity $2l+1$, and its trace over $l \le 2$ is $0+6+30 = 36$; B is larger because the $l = 1$ multiplet occurs in two shells ($n = 2$ and $n = 3$). Its value also depends on how each shell is labelled (see the scope section below).

The inner sum has the closed form

$$\sum_{l=0}^{n-1} (2l+1)\, l(l+1) = \frac{n^2(n^2-1)}{2}, \tag{3}$$

so the per-shell average of the $SO(3)$ Casimir equals the $SO(4)$ Casimir eigenvalue,

$$\langle C_3 \rangle_n = \frac{1}{n^2} \sum_{l=0}^{n-1} (2l+1)\, l(l+1) = \frac{n^2-1}{2} = C_4(n), \tag{4}$$

$n_{\max}$	$B(n_{\max})$	$N(n_{\max})$	B/N
1	0	1	0.000
2	6	5	1.200
3	42	14	3.000
4	162	30	5.400
5	462	55	8.400

TABLE I. $B/N = 3$ holds only at $n_{\max} = 3$, Eq. (6).

and

$$B(m) = \frac{m(m+1)(2m+1)(m+2)(m-1)}{20}. \quad (5)$$

The cutoff $n_{\max} = 3$

With $N(m) = \sum_{n=1}^{m} n^2 = m(m+1)(2m+1)/6$ the number of states through shell m,

$$\frac{B(m)}{N(m)} = \frac{3(m+2)(m-1)}{10}. \quad (6)$$

Setting $B/N = \dim(S^3) = 3$ gives $(m+2)(m-1) = 10$, i.e. $(m+4)(m-3) = 0$, whose only positive root is $m = 3$ (Table I). The criterion $B/N = \dim S^3$ is chosen; given it, the cutoff is computed. Equivalently, by Eq. (4), the degeneracy-weighted average of the $SO(4)$ Casimir over shells $1 \le n \le n_{\max}$ equals $\dim S^3$.

Verification. `tests/test_paper2_ingredients.py` recomputes $B(3) = 42$ term by term from Eq. (1), proves Eqs. (3)–(6) symbolically, and checks that $B/N = 3$ holds at $n_{\max} = 3$ and fails at $n_{\max} \in \{1, 2, 4, 5\}$, all in exact arithmetic.

S^3 specificity of the criterion. The analogous ratio built from eigenvalues alone is not specific to S^3. For the round sphere S^d with $\lambda_k = -k(k+d-1)$ and degeneracies g_k, the formal moment $B_{\text{formal}}(m) = \frac{1}{2}\sum_{k=0}^{m} g_k|\lambda_k|$ satisfies $B_{\text{formal}}(m)/N(m) = d$ exactly at $m = 2$ for every $d \ge 1$ [11] (`tests/test_paper2_corrections.py`). The B of Eq. (1) is the trace of the $SO(3)$ sub-Casimir over the $SO(4) \to SO(3)$ branching, which equals B_{formal} only for $d = 3$, where the diagonal $SO(3) \subset SO(4)$ has embedding index one.

F: a Dirichlet series of the shell degeneracy

The n-th scalar shell of S^3 has degeneracy n^2. Its Dirichlet series at $s = 4$ is

$$F \;=\; \sum_{n\ge1} \frac{n^2}{n^4} \;=\; \zeta_R(2) \;=\; \frac{\pi^2}{6} \;=\; 1.6449\ldots \quad (7)$$

The exponent $s = 4$ is chosen; it equals the maximum degree $d_{\max}$ of the packing construction (Paper 0) and the dimension of the ambient $\mathbb{R}^4$. The same number is half the spectral zeta function of the circle of circumference 2π at $s = 1$: $\zeta_{S^1}(s) = 2\sum_{k\ge1} k^{-2s}$, so $F = \frac{1}{2}\zeta_{S^1}(1)$, the $k > 0$ modes. Either reading is a choice of which spectral series to evaluate and where.

Δ: a reciprocal degeneracy

Two closed forms give $\Delta = 1/40$ at the cutoff. The first combines the scalar eigenvalue of the last kept shell with the number of states below it,

$$\Delta = \frac{1}{|\lambda_3|\, N(2)} = \frac{1}{8 \times 5} = 1/40, \quad (8)$$

with $\lambda_n = -(n^2 - 1)$. The second is the reciprocal of a Dirac degeneracy: the Camporesi–Higuchi spectrum on unit S^3 has $|\lambda_k^{\mathrm{D}}| = k + 3/2$ with multiplicity $g_k^{\mathrm{D}} = 2(k+1)(k+2)$, $k = 0, 1, \ldots$, and

$$\Delta^{-1} \;=\; g_3^{\mathrm{D}} \;=\; 2 \cdot 4 \cdot 5 \;=\; 40. \quad (9)$$

As functions of the cutoff m, $|\lambda_m| N(m-1)$ and g_m^{D} are different polynomials; among positive integers they agree only at $m = 3$. The Dirac index here is the Camporesi–Higuchi one: $k = 3$ is the fourth Dirac level ($|\lambda_3^{\mathrm{D}}| = 9/2$), so the shared integer 3 is shared across two labelling conventions. The physical significance of Δ is not established.

THE FORMULA AND THE CUBIC

Combining the three ingredients,

$$K = \pi(B + F - \Delta) = \pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right) = 137.036064\ldots, \quad (10)$$

which differs from $\alpha^{-1}_{\text{CODATA 2018}} = 137.035999084$ by 4.8×10^{-7} relative. The physical α approximately satisfies (to 8.8×10^{-8} relative through the cubic's root; an Observation) the relation $1/\alpha + \alpha^2 = K$, i.e. the depressed cubic

$$\alpha^3 - K\alpha + 1 = 0. \quad (11)$$

Its discriminant $4K^3 - 27$ is positive, so it has three real roots; we take the smallest positive one, the perturbative one, which gives $\alpha^{-1} = 137.036011$. Both K and the root lie above the measured value, the root closer by a factor of about five. With CODATA 2022 (137.035999177) the two relative differences are 4.76×10^{-7} and 8.75×10^{-8}.

Verification. `tests/test_paper2_corrections.py` recomputes the root ($1/\alpha = 137.036011$, above CODATA, within 10^{-7} relative).

The cubic is not selected by any structure we know. Every depressed cubic $s^3 - Ks + 1$ is the characteristic polynomial of a traceless 3×3 circulant with $bc = K/3$ and $b^3 + c^3 = -1$ (`tests/test_paper2_ingredients.py`), so that realization fixes nothing; the self-consistency form $1/\alpha + \alpha^2 = K$ is a choice.

Proposition 1 (The cubic's root is transcendental)**.** ***[SYMBOLIC]*** *The physical fine structure constant only* approximately *satisfies the cubic of Eq. (11) (to* 8.8×10^{-8} *relative, far outside the experimental uncertainty), so the cubic is not an exact law, and this statement concerns the cubic's smallest positive root* α_K*, not the physical* α*:* α_K *is algebraic over* $\mathbb{Q}(\pi)$ *and transcendental over* $\mathbb{Q}$*. Nothing follows for the physical* α*, and the combination rule remains an Observation.*

Proof. From $\alpha_K^3 - K\alpha_K + 1 = 0$ and $\alpha_K \neq 0$, $K = \alpha_K^2 + 1/\alpha_K$. If α_K were algebraic, so would be K. But $K = \pi(B + F - \Delta) = \frac{1679}{40}\pi + \frac{1}{6}\pi^3$ is a nonconstant polynomial in π with rational coefficients, hence transcendental (Lindemann). □ (The mechanism is specific: roots of polynomials with transcendental coefficients *can* be algebraic — e.g. $x^2 - \pi x + (\pi - 1) = (x - 1)(x - \pi + 1)$ has the root $x = 1$ — but that needs the π-dependence to cancel at the root. That cannot happen here: in $x^3 - Kx + 1$ the transcendental π enters only through the single coefficient K, the other coefficients being rational, so an algebraic root $x_0 \neq 0$ would force $K = (x_0^3 + 1)/x_0$ to be algebraic; in the example π enters two coefficients, and they cancel at $x = 1$.) (`tests/test_paper2_cubic_root_transcendence.py`.)

LOOK-ELSEWHERE COUNT

A formula built from π, 42 and $\pi^2/6$ that matches α^{-1} this closely could be a numerical accident. To estimate how often such matches arise, we ran a combinatorial search [10].

Search space

The search considers $\alpha^{-1} = T(A + B + C)$ and the cubic $\alpha^3 - T(A + B + C)\alpha + 1 = 0$, where

- A, B, C are drawn from a pool of spectral quantities: eigenvalues $|\lambda_n|$, degeneracies n^2 and $2l + 1$, Casimir values $l(l + 1)$, cumulative counts $N(n)$, their pairwise products up to 50, signed ratios of these, and the transcendentals $\zeta(2)$, $\zeta(3)$, $\zeta(4)$, $\ln 2$, π, π^2 (743 distinct term values);
- T ranges over 14 prefactors: π, 2π, π^2, $\pi/2$, 4π, $4\pi^2$, $1/(2\pi)$, 1, 2, 1/2, $\pi/4$, $\pi/6$, $1/\pi$, $\sqrt{\pi}$.

Threshold	Hits	Fraction
10^{-2}	1,283,592	6.7×10^{-4}
10^{-3}	125,027	6.5×10^{-5}
10^{-4}	14,476	7.5×10^{-6}
10^{-5}	682	3.5×10^{-7}
10^{-6}	37	1.9×10^{-8}
10^{-7}	11	5.7×10^{-9}
8.8×10^{-8}	10	5.2×10^{-9}

TABLE II. Formulas of the search matching α^{-1} within each relative threshold. The fraction is roughly linear in the threshold from 10^{-2} to 10^{-4}; below that the counts are small and include repeated values (one number reached by several formulas).

Both readings (linear and cubic) give $N = 1.92 \times 10^9$ formulas in all. The pool was built to contain this formula's own ingredients, so the count is conditional on that pool.

Results

Ten formulas match α^{-1} more closely than the cubic's root (8.81×10^{-8}), so this formula ranks eleventh, and the fraction at its precision is

$$p = \frac{10}{1.92 \times 10^9} = 5.2 \times 10^{-9}. \tag{12}$$

The expected number of chance matches at least this close in a search of this size is therefore

$$Np \;=\; 1.92 \times 10^9 \times 5.2 \times 10^{-9} \;\approx\; 10, \tag{13}$$

and ten are found. The ten take only three distinct values (the closest is $4\pi(5/49 + 14/15 + \pi^2)$, at 2.1×10^{-8}). At the precision of K itself (4.8×10^{-7}, the linear reading), 27 entries of the search match more closely, one of them this formula's own cubic reading [10].

Within this search, then, a match of this quality is what chance produces about ten times over; p is a fraction per formula, not a significance level. The search was designed and run after the formula was found, so it also does not account for the unregistered search that produced the formula, nor for the choice of pool. The count does not support the formula and does not refute it; it shows that precision alone carries no evidential weight here. (The full sweep is not regression-gated; the driver is preserved at `tests/paper2_support/phase2_combinatorial.py`.)

WHAT IS COMPUTED AND WHAT IS CHOSEN

Table III separates the steps from spectral data to the number that are computed from those that are chosen. The formula has no continuous fitted parameter, but it

has several discrete choices, and each of them enters the look-elsewhere count above as one draw from a pool of alternatives.

The sign check is the one place where the alternatives were scored. Of the eight patterns $\pi(\pm B \pm F \pm \Delta)$ at $n_{\max} = 3$, $(+,+,-)$ gives 4.8×10^{-7}, the next best $(+,+,+)$ gives 1.1×10^{-3}, and the rest are 7.4×10^{-2} or worse. Flipping the sign of Δ moves K by $2\pi\Delta$, which is 1.1×10^{-3} relative, and $\pi(42 + \pi^2/6)$ alone already lies within 5.8×10^{-4} of α^{-1}; the check therefore amounts to taking the sign of Δ that brings the sum closer, and it is not independent of the match itself.

WHERE THE INGREDIENTS LIVE, AND TWELVE ELIMINATED MECHANISMS

Proposition 2 (Three homes)**.** *The three ingredients B, F, Δ of the combination rule (an Observation; Eq.* (10)*) have the following spectral homes on S^3:*

- (B) *a finite truncated Casimir trace over the scalar Laplace–Beltrami harmonics at the cutoff $m = 3$, with weight $(2l+1)\,l(l+1)$, Eq. (1);*
- (F) *an infinite Dirichlet series of the scalar shell degeneracy at $s = 4$, $F = \sum_n n^2 \cdot n^{-4} = \zeta_R(2)$;*
- (Δ) *the reciprocal of a single-level Dirac degeneracy, $\Delta^{-1} = g_3^{\mathrm{D}} = 40$ (Camporesi–Higuchi index).*

They lie in two spectral sectors of S^3 (scalar and spinor) and are three different kinds of object (finite trace, infinite series, single-level multiplicity).

The proof is the closed forms of Eqs. (1), (7) and (9). [**OBSERVATION**] No single spectral construction on S^3 generating all three has been exhibited: twelve single-mechanism derivations have been eliminated (an empirical record, not a theorem; cf. the E6 scope remark of Paper 32).

Observation 1 (Cross-sector sum)**.** *The combination rule $K = \pi(B + F - \Delta)$ is a* cross-sector sum *of three ingredients with separate algebraic origins. That the sum comes within 4.8×10^{-7} of α^{-1}/π for K itself (to 8.8×10^{-8} through the cubic's root) is a numerical observation, not a consequence of Proposition 2; no common-generator mechanism for it is known.*

The twelve mechanisms

Table IV lists the mechanisms tested for a single spectral construction producing B, F and Δ together (or the combination K). Eight come from the structural program of Phases 4B–4I, one from Sprint A, and three from Sprint K-CC [12, 13]. The record of Phases 4B–4I counted the discrete and the continuum circle-fibre tests as two mechanisms (nine in all); the corpus count of twelve treats them as one, row 4. Elimination means the construction was computed and missed; it is an empirical record, not an exhaustion.

The three Dirac obstructions (row 8)

Obstruction 1 (to lifting B). The scalar Laplacian on S^3 has a zero mode at $(n = 1, l = 0)$, which forces the factor $(m-1)$ in Eq. (5). The Dirac operator has no zero mode ($|\lambda_k^{\mathrm{D}}| \geq 3/2$), and its cumulative traces,

$$\sum_{k=0}^{M} g_k^{\mathrm{D}} = \frac{2(M+1)(M+2)(M+3)}{3}, \tag{14}$$

$$\sum_{k=0}^{M} g_k^{\mathrm{Weyl}} = \frac{(M+1)(M+2)(M+3)}{3}, \tag{15}$$

carry no such factor. The single hit $|\lambda_2^{\mathrm{D}}| \cdot g_2^{\mathrm{Weyl}} = (7/2) \cdot 12 = 42$ follows from $(m-1)(m+2) = 10$ at $m = 3$, an elementary identity.

Obstruction 2 (to lifting F). The Dirac degeneracy splits as $g_m = 2m^2 + 2m$ in the integer label $m = k + 1$, so its Dirichlet series at $s = 4$ is

$$\sum_{m \geq 1} \frac{2m(m+1)}{m^4} \;=\; 2\,\zeta_R(2) + 2\,\zeta_R(3). \tag{16}$$

Conditional on $\zeta(3)/\zeta(2) \notin \mathbb{Q}$ (open: Apéry's theorem gives only $\zeta(3) \notin \mathbb{Q}$), no rational combination of these Dirac-sector series is a rational multiple of F; isolating F means projecting onto the m^2 part, i.e. returning to the scalar sector. In the Camporesi–Higuchi index the series is $\sum_{k \geq 0} 2(k+1)(k+2)/(k+3/2)^4 = \pi^2 - \pi^4/12$, which is unconditionally not a rational multiple of F (Lindemann).

Obstruction 3 (to unifying Δ with B or F). Under the S^1 action of the Hopf fibration (the Orduz S^1-equivariant decomposition of the spinor bundle), the 40 states of the Dirac level $k = 3$ split as $20+20$, by chirality or, equivalently, into half-integer and integer charge sectors. The decomposition produces neither B nor F, and the scalar factor $8 = |\lambda_3|$ of Eq. (8) is a scalar eigenvalue, not a spinor quantity, so the two factorizations of 40 do not combine.

The base spectrum and degeneracies are in `geovac/dirac_s3.py` (`tests/test_dirac_s3.py`, including $g_3^{\mathrm{D}} = 40$); the obstruction identities themselves were computed in archived sprint drivers and are recorded in `docs/dirac_s3_verdict.md`. The ingredients are also not Seeley–DeWitt coefficients of the perturbative spectral action (row 12); an earlier "Seeley–DeWitt cancellation theorem" supporting a stronger reading is withdrawn (2026-10-02; the exact S^3 heat traces give coefficient ratios $2, -1, 0, \ldots$, Paper 18).

Step	Status	Note
Weight $(2l+1)\,l(l+1)$ in B	chosen	over the rotation labelling of each shell
Criterion $B/N = \dim S^3$	chosen	
Cutoff $n_{\max} = 3$	computed	from the criterion, Eq. (6)
Values $B = 42$, $F = \pi^2/6$, $\Delta = 1/40$	computed	closed forms, Eqs. (1)–(9)
Series and point for F ($s = 4$, or $s = 1$ on S^1)	chosen	
Form of Δ	chosen	two forms agree at $m = 3$ only
Additive combination	chosen	
Signs $(+,+,-)$	chosen, checked	then the other seven sign patterns miss by $\geq 1.1 \times 10^{-3}$
Prefactor π	chosen	
Cubic $1/\alpha + \alpha^2 = K$	chosen	the circulant realization selects nothing
Root of the cubic	computed	the smallest positive root

TABLE III. Computed and chosen steps of the formula. "Chosen" means no construction we know fixes the step; it is the one that gives the match.

#	Mechanism	Where	Result
1	Spectral comparison of S^3 with the untwisted $S^1 \times S^2$	4B, α-A	zeta values, regularized determinants, truncated Casimir traces and heat-kernel coefficients of the two manifolds reproduce none of K, K/π, $B+F$, $B+F-\Delta$ better than 10^{-3}
2	Traces of higher Casimir powers on S^3	4B, α-B	$\sum_n n^2 c_2(n)^k$: only the $k=1$ trace gives a component ($42 = B$); none gives $\pi^2/6$ or K/π
3	Spectral invariants of the scalar label graph and its S^2 quotient at $n_{\max} = 3$	4B, α-D	trace, log-determinant, zeta, Cheeger constant and entropy of either Laplacian hit no target unless $(2l+1)l(l+1)$ is reinserted by hand
4	Circle-fibre traces, discrete (Fock weight) and continuum	4C α-E, 4D α-H	path-graph fibre zetas are rational under every weighting; continuum fibre traces give $\pi \cdot \mathbb{Q}$ at each asymptotic order (Jacobi inversion), never $\pi^2/6$
5	Spectral geometry of S^5	4E, α-I	volumes give integer powers of π, heat-kernel coefficients rationals, determinants ζ(odd) and $\log 2$; π^2 never appears beside a rational
6	Zeta-value combinations for Δ	4G, α-K	3,228 candidates at 50 digits: none within 10^{-25} of $1/40$
7	Standard-Model running (one-loop vacuum polarization) as the origin of Δ	4H, SM-A–F	one-loop QED running gives $\Delta(1/\alpha) = 1/40$ only at $\mu \approx 575$ MeV, no natural scale; $\sum N_c Q_f^2 = 8 = \lvert\lambda_3\rvert$ is charge-universal per generation, a coincidence; $1/40$ is no Chern number, η-invariant or Chern–Simons level of $SU(5)$, $Spin(10)$ or E_8 except by renaming
8	Lift of B, F, Δ to the Dirac spectrum on S^3	4I, D2–D4	obstructions 1–3 below
9	Connes–Chamseddine bulk spectral action on S^3	Sprint A, α-SP	the bulk expansion has all-positive Seeley–DeWitt terms on round S^3, so cannot give the pattern $(+,+,-)$
10	A cutoff Λ with $\mathrm{Tr}\, e^{-D^2/\Lambda^2} = K/\pi$, identified by integer relation	K-CC (a)	PSLQ of $\Lambda = 3.7102454\ldots$ at 100 digits against a 244-element basis: no relation
11	A natural Λ from S^3 spectral data	K-CC (b)	only tautological hits: every target has one real root Λ
12	B, F, Δ as three terms of one heat-kernel expansion	K-CC (c)	the Seeley–DeWitt coefficients of D^2 on unit S^3 lie in $\sqrt{\pi}\,\mathbb{Q}$, B is a finite trace of a different operator and Δ^{-1} a single-level count, so no one expansion contains all three

TABLE IV. The twelve eliminated mechanisms. "Where" names the phase or sprint and its track. Rows 1–8 are from the record of Phases 4B–4I, row 9 from Sprint A (2026-04-18), rows 10–12 from Sprint K-CC (2026-05-03) [12, 13].

SCOPE OF A HOPF-FIBRATION READING

Earlier versions of this paper read the three ingredients as data of the three spaces of the Hopf fibration $S^1 \to S^3 \to S^2$ [5]: B on the base, F on the fibre, Δ on the total space. Two measured facts limit any such reading.

The labelling matters. B is computed over the rotation grouping of each shell (orbits of $SO(3)$, labelled by l). Grouping each shell instead by the chiral $SU(2)$ factor whose torus is the Hopf fibre gives a Casimir trace of exactly $B/2 = 21$ at the same cutoff, because $\mathrm{Tr}\, L^2 = \mathrm{Tr}\, A^2$ on each hydrogen shell. The degeneracy n^2, and hence F, does not depend on the grouping (`tests/test_paper25_hopf_grouping.py`; record `docs/predictions/2026-10-31_hopf_grouping.md`). So $B = 42$ is attached to the rotation labelling, not to the fibration.

The fibration is a one-centre structure. The Hopf fibre is generated by $L_z + A_z$. With a second nucleus on the z axis, the commutant of its potential inside $\mathfrak{so}(4)$ is spanned by L_z alone, and the fibre generator is broken at relative size 0.38 (`tests/test_atlas_so4_commutant.py`; record `docs/predictions/2026-10-30_so4_commutant_door11.md`). Any reading of K through the fibration therefore concerns a single Coulomb centre.

COMPARISON

Wyler's formula [8] gives $\alpha^{-1} = 137.03608$, about 6×10^{-7} from the measured value, comparable to K here. Both formulas rest on a combination rule without a mechanism. What this paper adds is bookkeeping: the computed-versus-chosen ledger of Table III, a look-elsewhere count (Table II), and a record of the mechanisms that were tested and failed (Table IV).

CONCLUSION

The number $K = \pi(42 + \pi^2/6 - 1/40)$ lies within 4.8×10^{-7} of α^{-1}, and the smallest positive root of $\alpha^3 - K\alpha + 1 = 0$ within 8.8×10^{-8} of α. The ingredients are spectral data of S^3: a Casimir trace over the scalar harmonics of the first three shells (the only cutoff with $B/N = \dim S^3$), a Dirichlet series of the shell degeneracy, and a reciprocal Dirac degeneracy. The combination rule $K = \pi(B + F - \Delta)$ is a numerical Observation. Its weights, signs, prefactor and the cubic are chosen (Table III). In a search of 1.92×10^9 formulas of the same kind the expected chance count at this precision is $Np \approx 10$ ($p = 5.2 \times 10^{-9}$ per formula), and ten closer matches are found, so precision alone does not distinguish this formula. Twelve proposed single-construction mechanisms were tested and eliminated (Table IV). No derivation of the combination rule exists.

This work supersedes an earlier "symplectic impedance" approach [7] by the same author. The computational audit that checked the formula's arithmetic and ran the look-elsewhere search is available as supplementary material [10]. Material cut from earlier versions of this paper (interpretive readings of the ingredients, the circulant analysis, a spectral-determinant near-miss, and the residual after the cubic) is preserved in the project repository at `docs/chronicles/paper2_narrowing_cut.md`. Computational implementation and documentation drafting were performed with the assistance of large language models (Anthropic Claude) under the author's scientific direction.

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